

Power BI Evidence

Task 7 overview: connection, model and dashboard

The Power BI Desktop was then connected to the SQL Server database CzechBank_DW (localhost) via Import mode. The fact and dimension tables were created by Member 2 and loaded by Member 3 through an ETL process, with mart.MonthlyTransactionPerformance and mart.DistrictLoanPerformance being the two additional pre-aggregated data-mart tables. The dashboard pages utilize mart tables which ensure that the visuals remain quick with more than 1,056,320 transaction rows whereas the fact loan and its dimensions are used for loan analysis.

Eight DAX measures were then created (Table 1) and there are three pages in the report: Executive Summary, Trend Analysis and Interactive Analysis. Drill down and roll up features have been enabled for the Region/District visualization. The totals have been reconciled against the warehouse via SQL (Figures 17 and 18).

1. Connection / Navigator

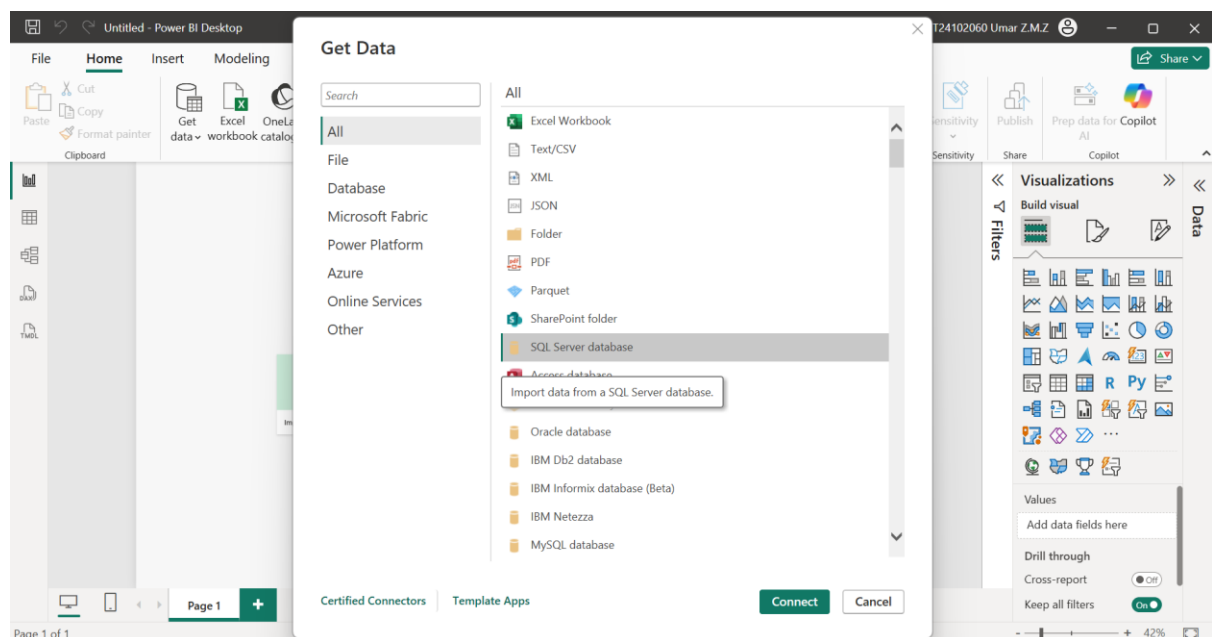


Figure 1: Get Data in Power BI Desktop with the SQL Server database connector selected

SQL Server database

Server ⓘ

localhost

Database (optional)

CzechBank_DW

Data Connectivity mode ⓘ

☒ Import
 ☐ DirectQuery

▶ Advanced options

OK

Cancel

Figure 2: Connection to server localhost, database CzechBank_DW, using Import mode

Navigator

Display Options ▾

card

client

☒ DimAccount

☒ DimClient

☒ DimDate

☒ DimDistrict

☒ DimLoanStatus

☒ DimOperation

☒ DimTransactionType

disp

district

☒ FactLoan

☒ FactTransaction

loan

☒ mart.DistrictLoanPerformance

☒ mart.MonthlyTransactionPerformance

order

stg.account

stg.account clean

mart.MonthlyTransactionPerformance

Year	MonthNumber	MonthName	Quarter	YearMonthLabel	TotalTransa
1993	1	January	1	1993-01	
1993	2	February	1	1993-02	
1993	3	March	1	1993-03	
1993	4	April	2	1993-04	
1993	5	May	2	1993-05	
1993	6	June	2	1993-06	
1993	7	July	3	1993-07	
1993	8	August	3	1993-08	
1993	9	September	3	1993-09	
1993	10	October	4	1993-10	
1993	11	November	4	1993-11	
1993	12	December	4	1993-12	
1994	1	January	1	1994-01	
1994	2	February	1	1994-02	
1994	3	March	1	1994-03	
1994	4	April	2	1994-04	
1994	5	May	2	1994-05	
1994	6	June	2	1994-06	
1994	7	July	3	1994-07	
1994	8	August	3	1994-08	
1994	9	September	3	1994-09	
1994	10	October	4	1994-10	

Select Related Tables

Load

Transform Data

Cancel

Figure 3: Navigator showing the dimension, fact and data-mart tables selected for loading.

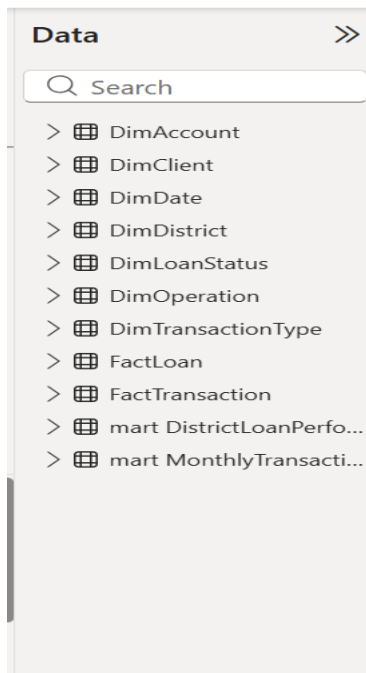


Figure 4: Data pane listing the tables loaded into the model

2. Model View

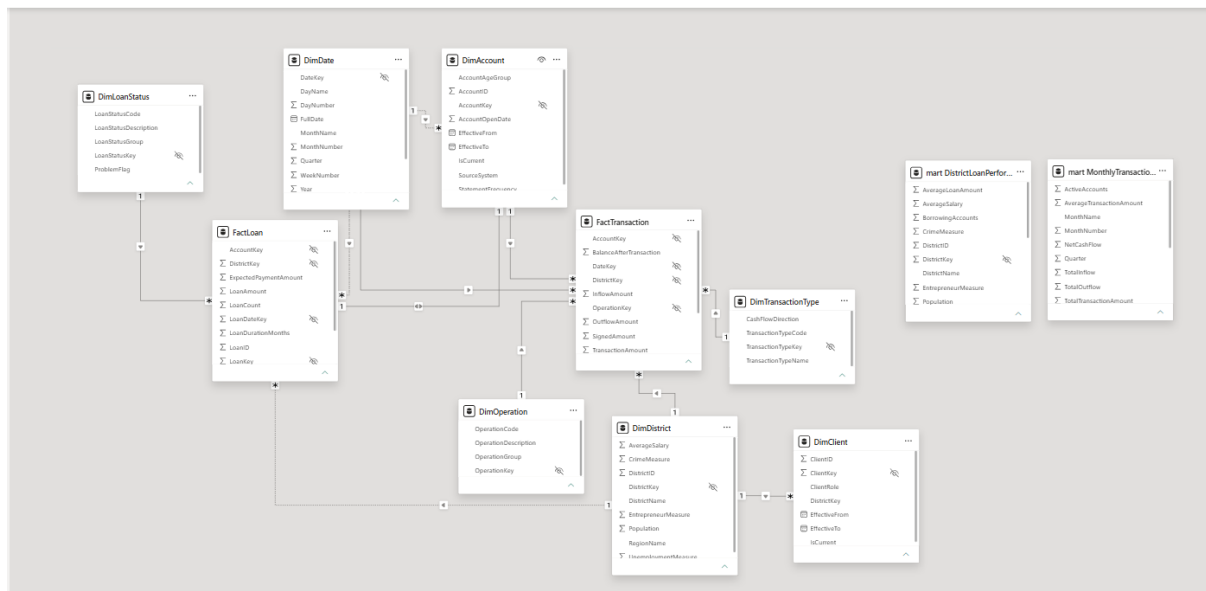


Figure 5

Model view. Fact tables connect to dimension tables through surrogate keys (mostly many-to-one with single-direction filtering); the two mart tables are standalone reporting tables

3. Measures List And Hierarchies for OLAP

mart MonthlyTransactionPerformance

 Total Transactions
1 Total Transactions = SUM('mart MonthlyTransactionPerformance'[TotalTransactions])
 Total Inflow 
1 Total Inflow = SUM('mart MonthlyTransactionPerformance'[TotalInflow])
 Total Outflow
1 Total Outflow = SUM('mart MonthlyTransactionPerformance'[TotalOutflow])
 Net Cash Flow 
1 Net Cash Flow = SUM('mart MonthlyTransactionPerformance'[NetCashFlow])

Figure 6: Measures defined in mart MonthlyTransactionPerformance

mart DistrictLoanPerformance

 Total Loans
1 Total Loans = SUM('mart DistrictLoanPerformance'[TotalLoans])
 Total Loan Amount
1 Total Loan Amount = SUM('mart DistrictLoanPerformance'[TotalLoanAmount])
 Problem Loans
1 Problem Loans = SUM('mart DistrictLoanPerformance'[ProblemLoans])
 Problem Loan Rate
1 Problem Loan Rate = DIVIDE([Problem Loans], [Total Loans], 0)

Figure 7: Measures defined in mart DistrictLoanPerformance

Changed the format to percentage for Problem Loan Rate Measure

TABLE: Measure Definitions

MEASURE	DAX FORMULA	Purpose and Where Used
Total Transactions	SUM('mart MonthlyTransactionPerformance'[TotalTransactions])	Overall transaction volume. Page 1

		card, Page 2 line chart.
Total Inflow	SUM('mart MonthlyTransactionPerformance'[TotalInflow])	Money received into accounts. Page 1 card, Page 2 columns.
Total Outflow	SUM('mart MonthlyTransactionPerformance'[TotalOutflow])	Money paid out of accounts. Page 1 card, Page 2 columns.
Net Cash Flow	SUM('mart MonthlyTransactionPerformance'[NetCashFlow])	Inflow minus outflow; liquidity indicator. Page 1 card.
Total Loans	SUM('mart DistrictLoanPerformance'[TotalLoans])	Number of loans issued. Page 1 card, Page 3 table.
Total Loan Amount	SUM('mart DistrictLoanPerformance'[TotalLoanAmount])	Value of loans issued (exposure). Page 3 bar chart.
Problem Loans	SUM('mart DistrictLoanPerformance'[ProblemLoans])	Loans with repayment problems (status B and D). Page 1 card, Page 3 table.
Problem Loan Rate	DIVIDE([Problem Loans], [Total Loans], 0)	Share of loans in a problem status; formatted as a percentage. Page 1 card, Page 3 table.

Hierarchies for OLAP

Hierarchies allow users to drill down and roll up through levels of detail. The following hierarchies support the OLAP analysis on Pages 2 and 3 (Table 2). The Region / District hierarchy is used for drill-down and roll-up in Figures 13 to 16.

TABLE: Hierarchies defined in the model.

Hierarchy	Table	Levels
Time	DimDate	Year → Quarter → MonthName
Geography	DimDistrict	RegionName → DistrictName

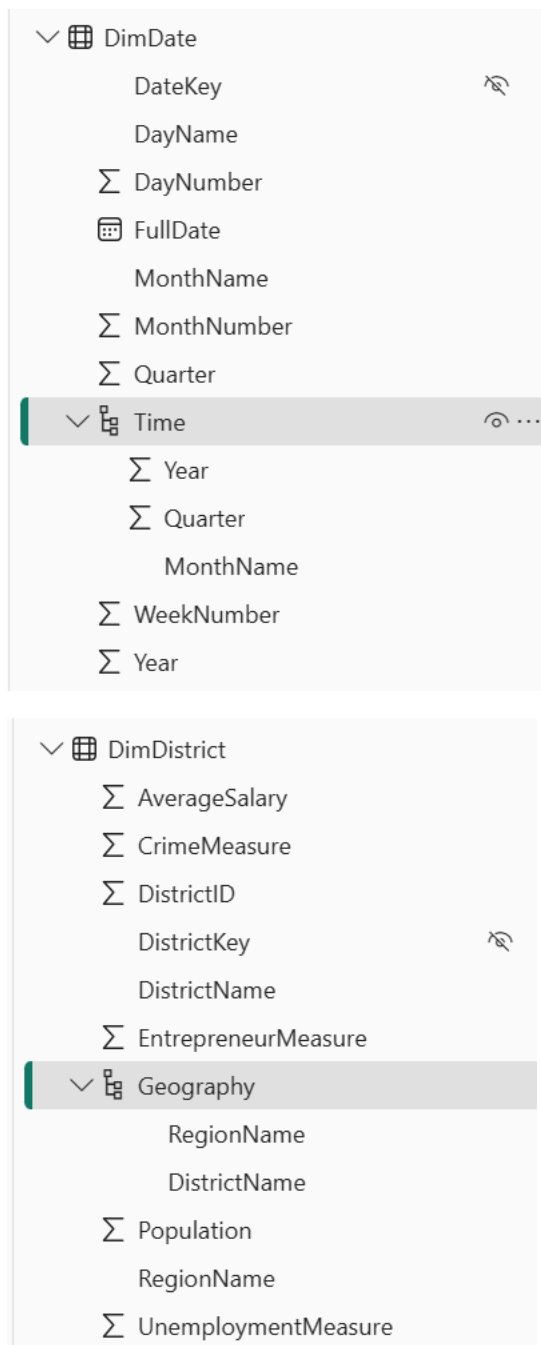


Figure 8: Evidence of Hierarchy

4. Three Pages

Build Page 1 - Executive Summary with KPI cards and overall performance indicators

Retail Banking Performance – Executive Summary

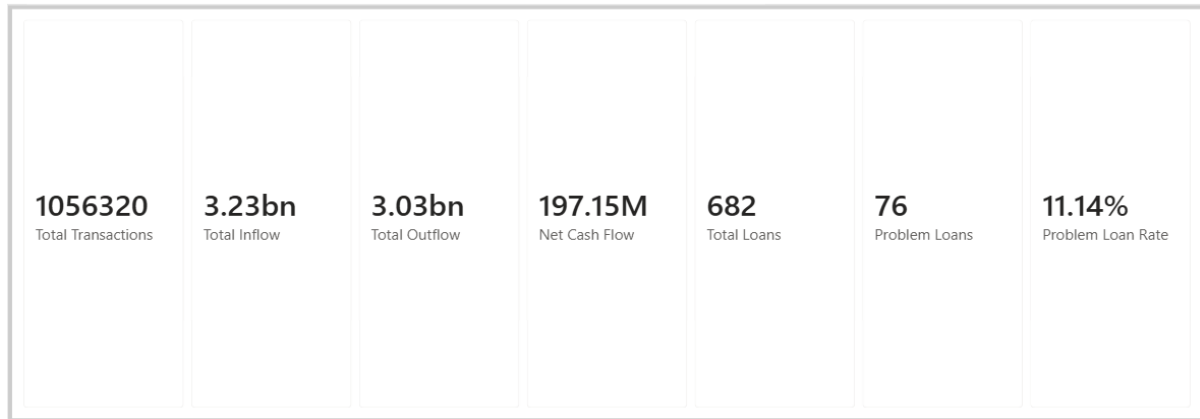


Figure 9: Page 1 – Executive Summary: KPI cards for Total Transactions (1,056,320), Total Inflow (3.23bn), Total Outflow (3.03bn), Net Cash Flow (197.15M), Total Loans (682), Problem Loans (76) and Problem Loan Rate (11.14%).

Build Page 2 - Trend Analysis with time-based line charts and comparisons.

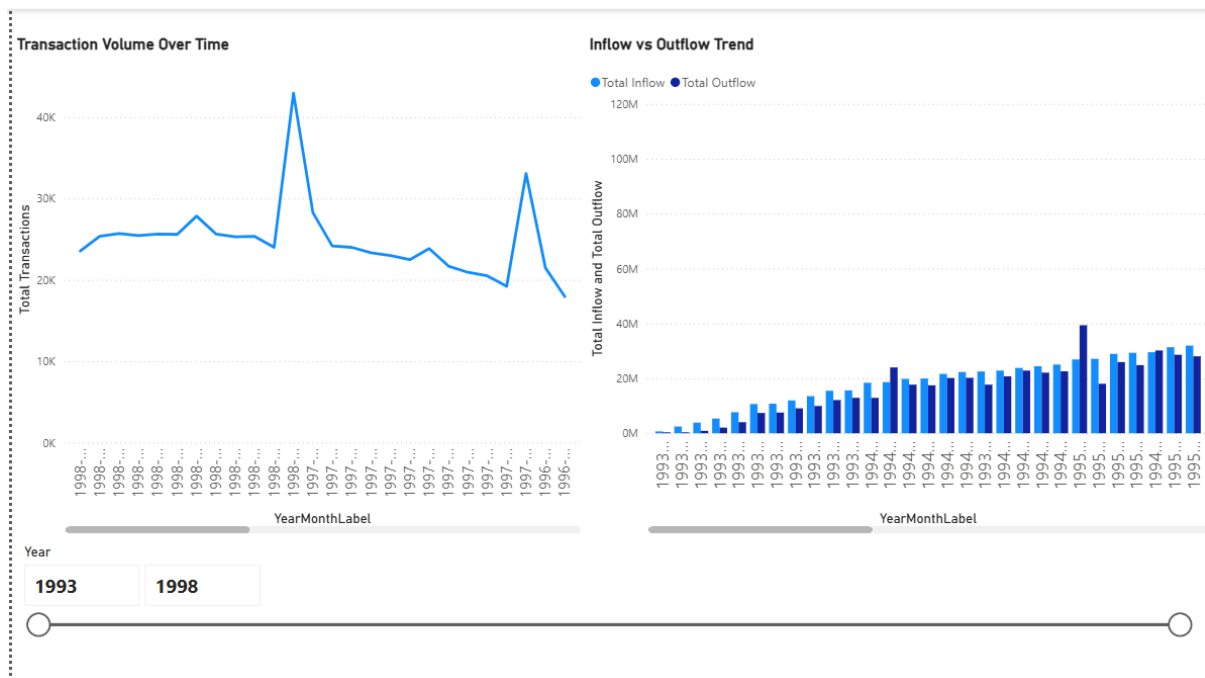


Figure 10: Page 2 – Trend Analysis: monthly transaction volume (line chart) and inflow vs outflow (clustered columns), with a Year slicer

Build Page 3 - Interactive Analysis with filters, slicers, drill-down and roll-up.

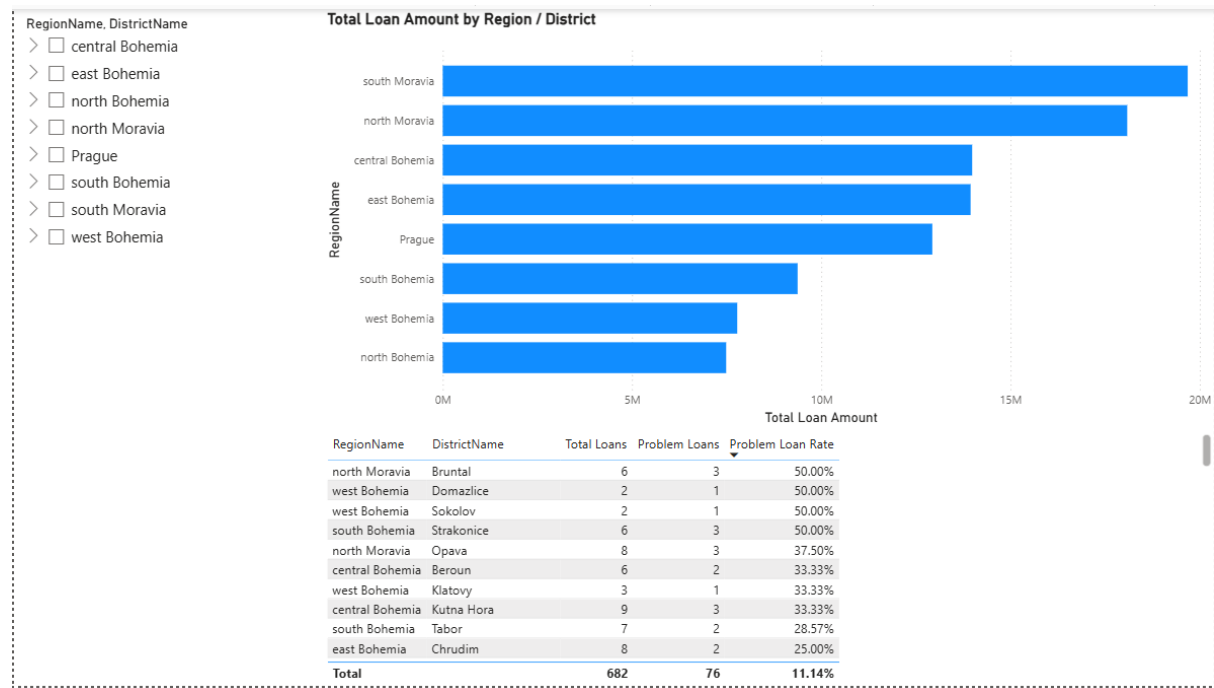


Figure 11: Page 3 – Interactive Analysis: Region / District slicer, drill-down bar chart of Total Loan Amount, and detail table of loans, problem loans and problem loan rate.

Slicers Applied

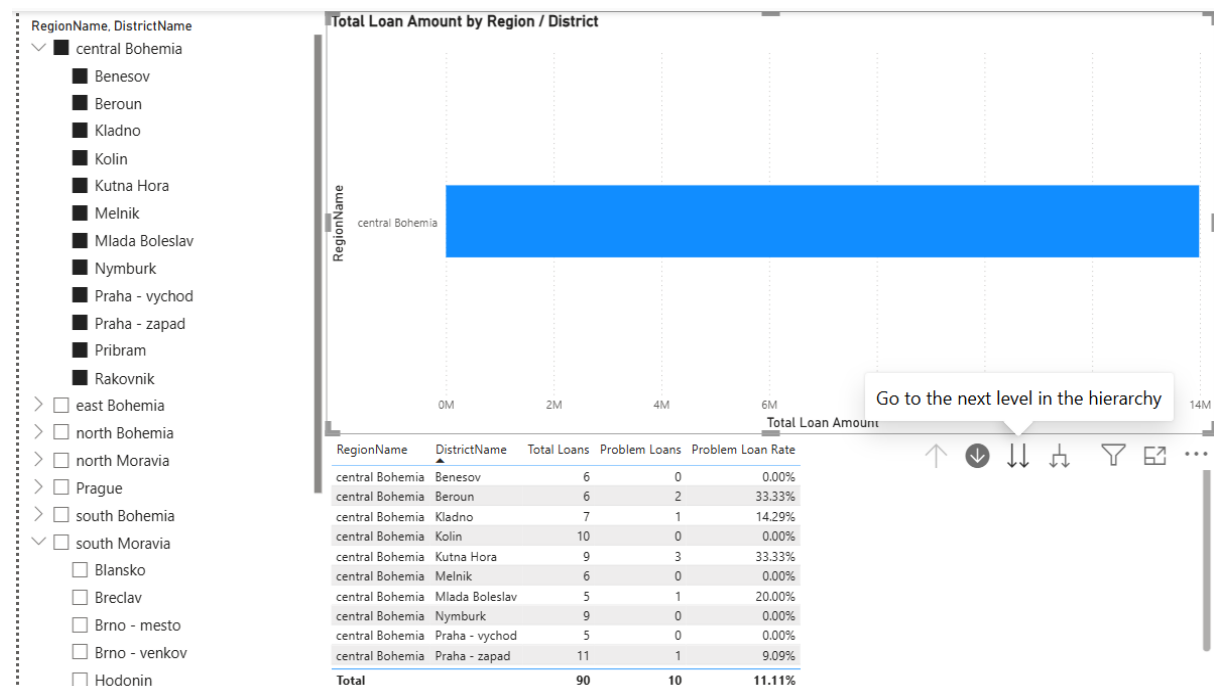


Figure 12: Slicer applied: central Bohemia selected, so the chart and table filter to that region (90 loans, 10 problem loans, 11.11%).

Drill-mode

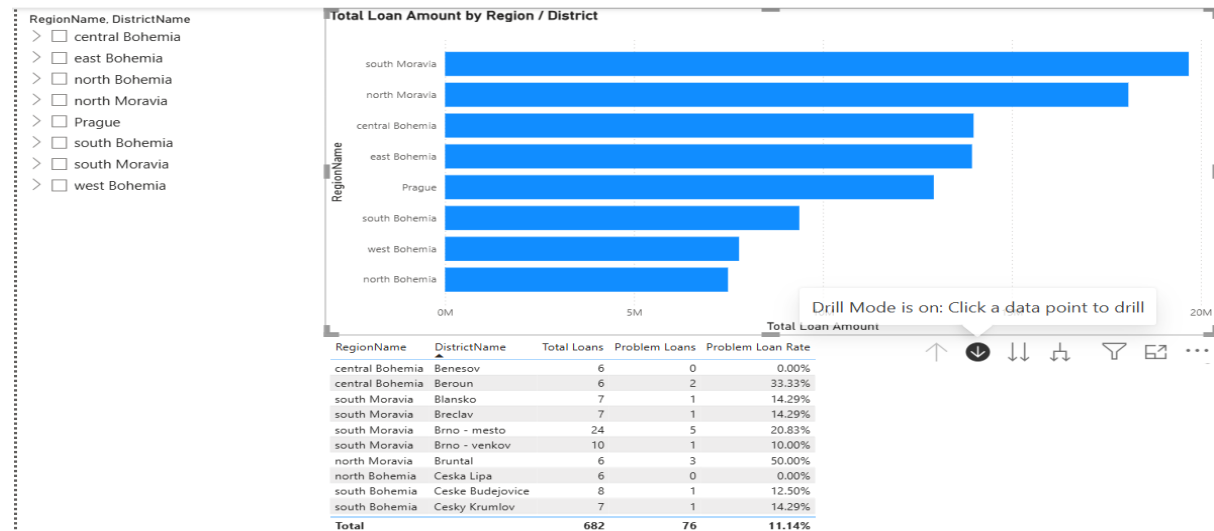


Figure 13: Drill mode switched on in the chart header.

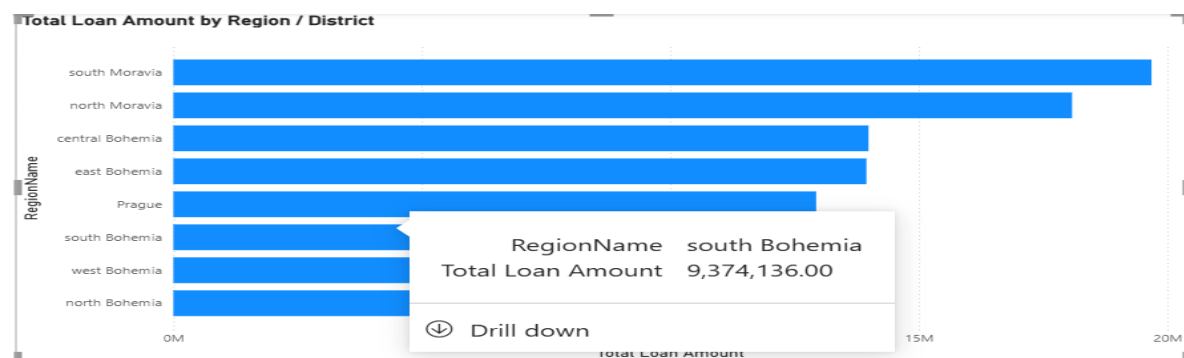


Figure 14: Drill down: selecting the south Bohemia bar and choosing Drill down.

Roll-up

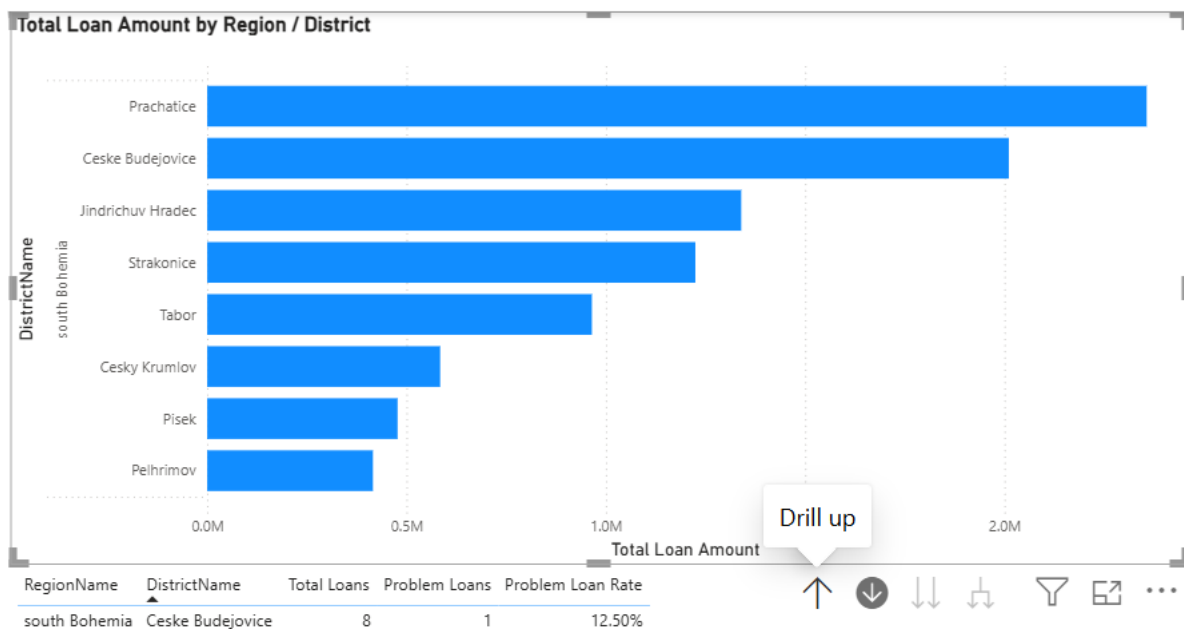


Figure 15: Result of the drill-down: districts within south Bohemia, with the Drill up button available.

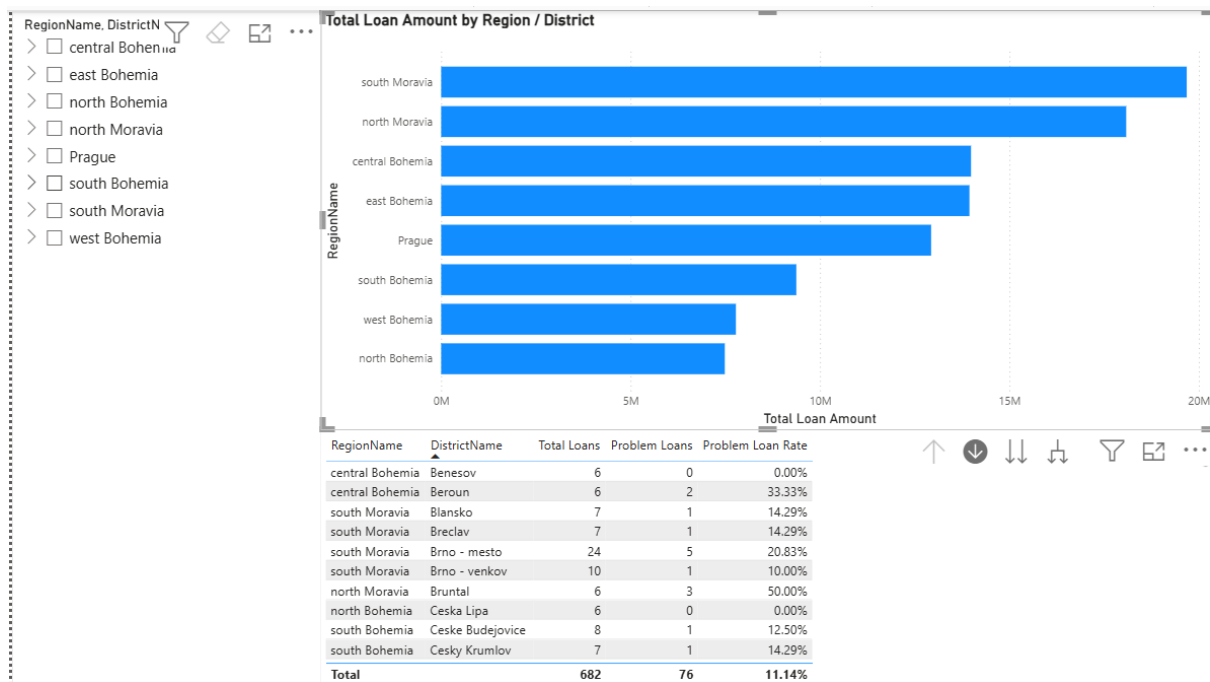


Figure 16: Roll-up: after Drill up the chart returns to region level.

5. Power BI totals match SQL warehouse totals

Total Transactions	Total Loans	Problem Loans
1056320	682	76

Figure 17: Power BI table showing 1,056,320 transactions, 682 loans and 76 problem loans.

SQLQuery1....0R\HP (60)*			
1	SELECT		
2	(SELECT COUNT(*) FROM dbo.FactTransaction) AS Transactions,		
3	(SELECT COUNT(*) FROM dbo.FactLoan) AS Loans,		
4	(SELECT SUM(ProblemLoans) FROM mart.DistrictLoanPerformance) AS ProblemLoans;		
110 % No issues found			
Results Messages			
	Transactions	Loans	ProblemLoans
1	1056320	682	76

Figure 18: SQL query on CzechBank_DW returning the same totals.

Both tools return the same values, so the dashboard totals reconcile to the warehouse and data mart

Insights and Recommendations and Decision Supports

Seven insights were derived from the dashboard pages

1. The bank is cash-positive overall, but the margin is thin

Evidence (Page 1): Inflow is 3.23bn while outflow is 3.03bn, resulting in net cash flow of 197.15M, which is just around 6% of inflows.

Recommendation: Monitor net cash flow on a monthly basis as one of key liquidity indicators. Establish internal minimum level of net cash flow as a percentage of inflows.

Decision Supported: Liquidity monitoring and pricing: establishes minimum net cash flow margin and guides deposits and pricing decisions.

2. The transactions have been increasing at an extremely rapid rate during this period.

Evidence (Page 2 line chart): Transactions increase from 177 per month in January 1993 to around 25,000 per month by the end of 1998. Total transactions equal 1,056,320 (Page 1).

Recommendation: Plan for capacity, support staff, and fraud control based on continuing volume increases. Utilize increased customer base to cross-sell.

Decision supported: Capacity planning and resources allocation: informs staffing, capacity and fraud monitoring budgeting.

3. January is the most active month.

Evidence (Page 2 line graph): The highest peaks are found in January. Jan 1998 has a value of 42,940 transactions, while Jan 1997 has a value of 33,060.

Recommendation: Staff and system should be expanded before January comes around, and maintenance can be done during February – April..

Decision supported: Operations Scheduling: helps in creating staff rosters and maintenance windows.

4. The cash flow deficits occur only in January and June, but those in January are increasing.

Evidence (page 2 inflows vs outflows columns): There are 10 instances where outflow is higher than inflow in 72 months, but all 10 instances occur in either January or June. The January deficit gap keeps widening every year, from an estimated -5.4M in 1994 to an estimated -33.5M in 1998.

Recommendation: Prepare a liquidity cushion before every January and June based on the last year's deficit gap (at least 35M in January 1999). Find the cause of the huge January outflow.

Decision supported: Treasury and cash reserve planning: sets the size of the liquidity buffer held before January and June.

5. There are only a few locations with loan exposure.

Evidences (Page 3 bar chart): Hl.m. Praha has the highest exposure of around 12.9M, 3 times greater than the second highest district (Brno - mesto at around 4.0M). On a region basis, south Moravia (around 19.7M) and north Moravia (around 18.1M) have the highest exposures.

Recommendation: Establish maximum exposure thresholds on district and regional basis and monitor on a monthly basis. Start with south and north Moravia portfolios first.

Decision supported: Portfolio risk limits: supports maximum exposure limits per district and region.

6. The problem loan risks show wide regional variations, and small districts may give misleading results.

Evidence (Page 3 table and slicer): The average figure stands at 11.14% (76 out of 682 loans). Choose any region using the slicer, and the total in the table changes accordingly. The figure for north Moravia is 15.4% (18 out of 117), while north Bohemia

has only 1.6% (1 out of 61). From the table, Bruntal comes in at 50% but on just six loans.

Recommendation: Rank or flag districts only if they have ten or more loans. Credit check priority goes to north Moravia because it has the highest problem loans in the regions.

Decision supported: Regional credit policy: directs credit review effort and avoids decisions based on very small samples.

7. A few high-exposure areas require a watchlist, whereas Prague is relatively secure.

Evidence (Table on page 3, ordered according to Problem Loan Percentage): In Brno - mesto, 24 loans include 5 problem loans, representing 20.83% of total loans, which is almost twice the bank average. Svitavy has 3 problem loans among 12 total loans (25%) and Ostrava - mesto has 3 out of 20 (15%). Prague, despite the highest exposure, has only 7 out of 84 (8.33%) problem loans, below the bank average.

Recommendation: Create a watchlist for Brno - mesto, Svitavy and Ostrava - mesto with a stricter loan approval process. Consider Prague as a large but relatively low-risk exposure.

Decision supported: Targeted credit controls: supports watchlists and tighter approval checks where risk is highest, while treating Prague as a large but lower-risk exposure.

Demo plan

TABLE: Four-Minute Demo Flow

Time	Page	What to show	Key message
0:00–0:30	Page 1	KPI cards; Figures 17–18 reconciliation	Bank is cash-positive (197.15M net) and dashboard totals match the warehouse.
0:30–1:30	Page 2	Line chart and inflow vs outflow (full 1993–1998 range)	Volume grows strongly, January peaks, deficits occur only in January and June and are widening.
1:30–3:00	Page 3	Live: pick a region in the slicer, drill down to districts, drill up, sort table by rate	Exposure is concentrated and problem-loan risk differs sharply by region.
3:00–4:00	Insights	Insights 4, 5 and 7	Recommendations: liquidity buffer, exposure limits and a credit watchlist.

Acknowledgements

The dimensional warehouse (dimensions, fact tables, keys, relationships and database backup) was designed and built by Member 2. The ETL process, data quality validation and the Retail Banking Performance Data Mart were implemented by Member 3. This section (Tasks 7 and 8) builds on those outputs and does not repeat their transformations.

AI-use acknowledgement

AI tools were used as a supporting tool for learning and development: to explain Power BI features, suggest page layouts and DAX patterns, and help with the insights format. All data connections, measures, dashboards, screenshots and final wording were produced by me, and every figure quoted in the insights was checked in Power BI and reconciled to the SQL warehouse totals before inclusion.